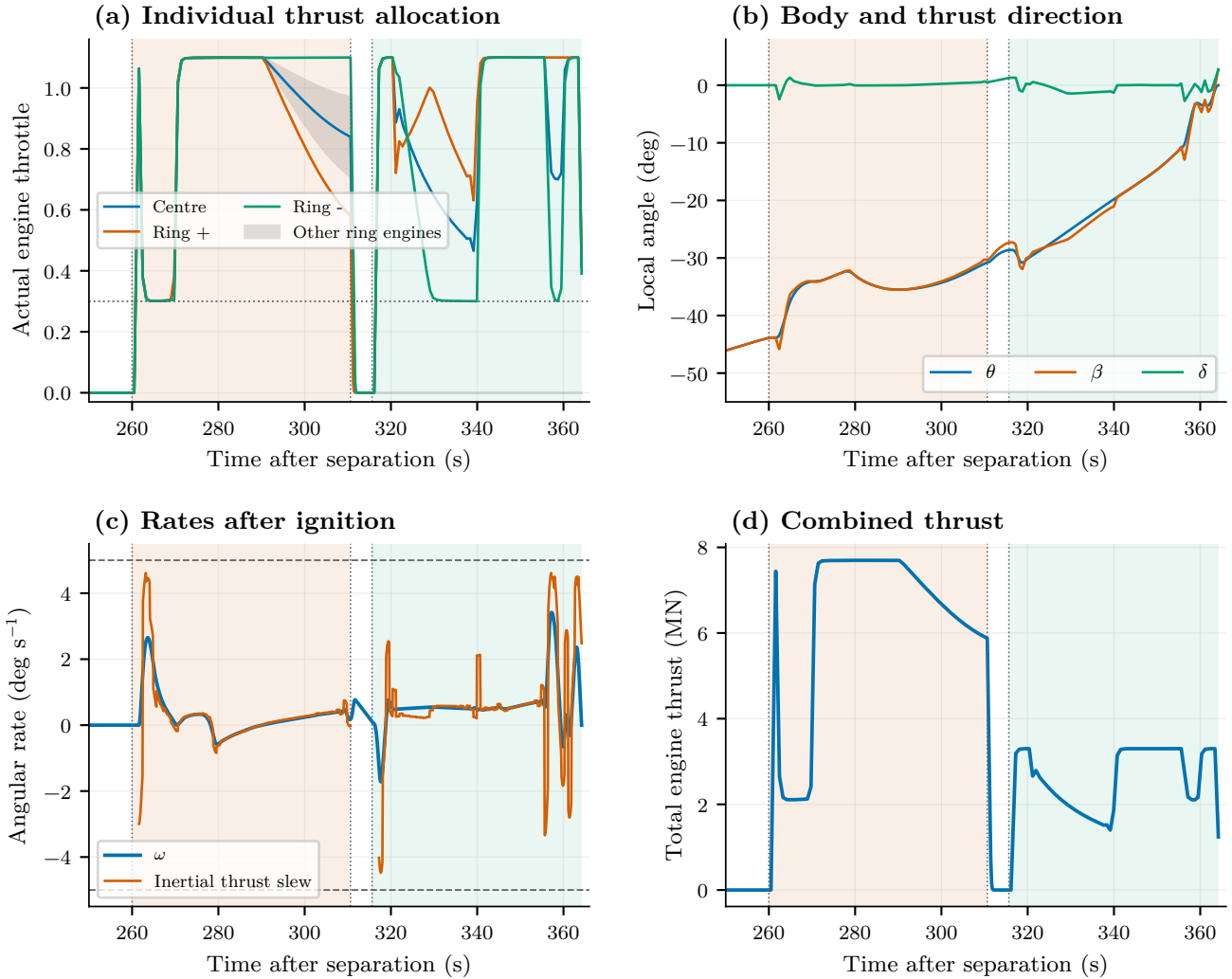

(a) Individual thrust allocation
Actual engine throttle
Time after separation (s)
Centre
Ring +
Ring -
Other ring engines
(b) Body and thrust direction
Local angle (deg)
Time after separation (s)
θ
β
δ
(c) Rates after ignition
Angular rate (deg s⁻¹)
Time after separation (s)
ω
Inertial thrust slew
(d) Combined thrust
Total engine thrust (MN)
Time after separation (s)